Palatial Pools Product Knowledge Hub
Proprietary Technology Technical Specification
Palatial Pool Services Pty Ltd
Classification: Proprietary & Confidential
Engineered by: Kieran Goldsmith

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Introduction

The **Palatial Pools Product Knowledge Hub** (PKH) is an advanced, engineering-grade digital library designed to serve technicians, engineers, and procurement professionals within the pool services industry. It provides immediate access to over 2,800 verified pool industry products, with comprehensive technical specifications, installation documentation, warranty details, compatibility matrices, and real-time pricing data. The PKH enables efficient product identification, compatibility assessment, and sourcing, thereby optimizing field operations and supply chain management.

This document provides a detailed technical exposition of the PKH system architecture, core technologies, data models, algorithms, security controls, and integration methods. It is intended for technical evaluators, system integrators, and stakeholders requiring an in-depth understanding of the technology foundation and operational capabilities.

System Overview

The Product Knowledge Hub is a multi-tiered information platform that consolidates heterogeneous data sources into a unified, searchable, and secure repository. It supports the following core functions:

- **Product Catalog Management:** Aggregation and normalization of product data from multiple manufacturers and suppliers.
- **Search and Query Engine:** Advanced keyword, category, and semantic search capabilities.
- **Compatibility Analysis:** Automated cross-referencing of products against customer equipment inventories to recommend compatible replacements.
- **Pricing and Availability:** Real-time pricing extracted via supplier APIs.
- **Documentation Access:** Secure access to manufacturer technical documents, installation guides, and warranty terms.
- **User Access Control:** Role-based access with strict data governance, ensuring compliance with Australian data sovereignty laws and corporate confidentiality policies.

Data Model and Product Catalog Architecture

Data Sources and Ingestion Pipeline

Product data originates from verified manufacturer feeds, supplier APIs, and in-house technical documentation. Data ingestion employs an Extract-Transform-Load (ETL) pipeline designed for high data integrity and normalization.

- **Extract:** Scheduled batch and near-real-time pull from RESTful APIs, FTP drops, and email attachments.
- **Transform:** Data cleansing, de-duplication, standardization of nomenclature, units, and taxonomies.
- **Load:** Storage in a relational database optimized for querying complex product relationships.

Core Data Entities

Entity	Description
Product	Unique identifier (SKU/Part Number), brand, model, specifications (dimensions, materials, power)
Category	Hierarchical classification aligned with industry standards (e.g., AS 3634 for pool equipment)

```
| **Compatibility Map**| Relationships defining compatible or alternative
products                                          |
| **Documentation**    | Links to PDF, CAD files, installation guides tagged
by product                                        |
| **Pricing**          | Supplier-specific current price, currency, and
availability status                               |
| **Customer Equipment**| Records of installed products per customer for
compatibility checking                            |

### Product Schema (Simplified JSON Schema Example)

```json
{
  "productId": "string",
  "brand": "string",
  "model": "string",
  "category": "string",
  "specifications": {
    "dimensions": {"height": "number", "width": "number", "depth":
"number"},
    "material": "string",
    "powerRating": "number",
    "voltage": "number"
  },
  "compatibility": ["productId1", "productId2"],
  "documentationUrls": ["url1", "url2"],
  "pricing": {
    "supplierId": "string",
    "price": "number",
    "currency": "string",
    "lastUpdated": "ISO 8601 timestamp"
  }
}
```

Search and Recommendation Engine

Search Engine Architecture

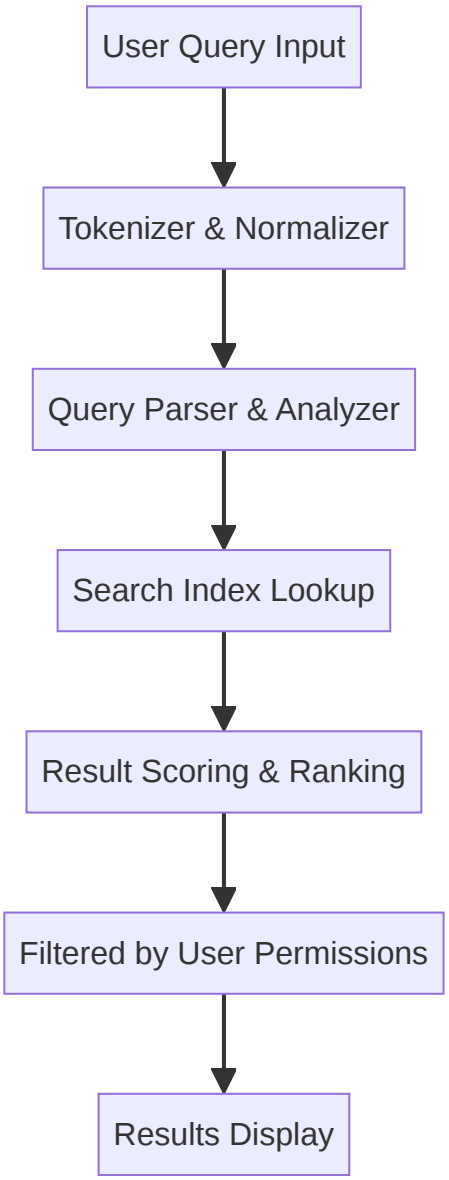
The search engine supports multiple query types:

- **Exact Match:** Part numbers, SKUs, and brand names.
```

- **Category Browsing:** Hierarchical navigation through categories and subcategories.
- **Natural Language Queries:** Keyword extraction and semantic matching using TF-IDF and word embeddings.

The system indexes product fields using Apache Lucene-based technology, enabling efficient full-text search with ranking based on relevance and usage frequency.

Query Processing Pipeline



Recommendation Engine

The recommendation module uses graph-based algorithms to identify compatible alternatives and replacements.

- **Compatibility Graph:** Nodes represent products; edges denote compatibility relationships.
- **Algorithm:** Modified breadth-first search (BFS) to traverse compatibility edges and rank alternatives by similarity scores weighted by specification proximity, manufacturer reliability, and availability.
- **Fallback Logic:** If exact product is discontinued or unavailable, suggest nearest compatible alternatives within defined tolerance thresholds.

Mathematically, similarity (S) between products (P_a) and (P_b) is computed as:

[S(P_a, P_b) = w_1 \cdot \text{spec_similarity}(P_a, P_b) + w_2 \cdot \text{brand_proximity}(P_a, P_b) + w_3 \cdot \text{availability_score}(P_b)]

where (w_i) are empirically derived weights.

Security Architecture

Data Integrity and Authenticity

- All product data is sourced from verified manufacturer feeds with digital signatures or secure API keys to ensure authenticity.
- ETL pipelines validate data integrity using SHA-256 checksums.

Access Control

- Role-Based Access Control (RBAC) implemented via OAuth 2.0 and OpenID Connect for user authentication and authorization.
- Pricing data is restricted to authorised personnel only, enforced by access control middleware.

Data Transmission and Storage

- All network communications enforce TLS 1.3 encryption with AES-256 symmetric cipher suites, adhering to Australian Signals Directorate (ASD) Information Security Manual (ISM) guidelines.
- Data at rest is encrypted using AES-256 with per-tenant key management.

Audit and Logging

- Immutable audit logs capture all access and modification events, stored in a secure, tamper-evident ledger.
- Regular security assessments and penetration testing are conducted in accordance with ISO/IEC 27001 standards.

Integration and API Specifications

API Overview

The Product Knowledge Hub exposes RESTful APIs for integration with third-party systems including ERP, CRM, and field service management platforms.

- **Base URL:** `https://api.palatialpools.com/v1/`
- **Authentication:** OAuth 2.0 Bearer Tokens
- **Data Format:** JSON (RFC 8259 compliant)

Key Endpoints

Endpoint	Method	Description
`/products`	GET	Retrieve list of products, supports filters
`/products/{id}`	GET	Retrieve detailed product information
`/products/{id}/compatibility`	GET	List compatible or alternative products
`/pricing/{productId}`	GET	Retrieve current pricing for a product
`/documentation/{productId}`	GET	Access technical documents linked to a product

Request and Response Example

Request: Retrieve product details

```
 GET /products/PLT-12345 HTTP/1.1
Host: api.palatialpools.com
Authorization: Bearer <token>
Accept: application/json
```

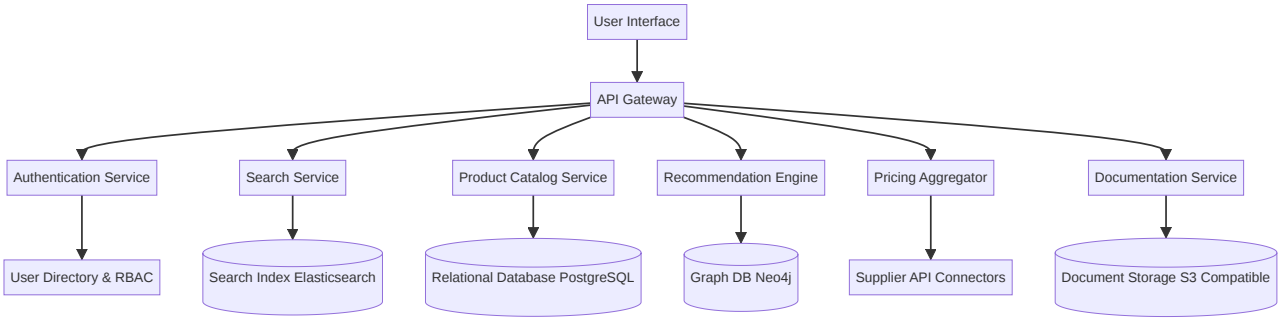
Response:

```
{
  "productId": "PLT-12345",
  "brand": "AquaTech",
  "model": "ATX-900",
  "category": "Pool Pumps",
  "specifications": {
    "dimensions": {"height": 35, "width": 20, "depth": 20},
    "material": "PVC",
    "powerRating": 1.5,
    "voltage": 240
  },
  "compatibility": ["PLT-12346", "PLT-12347"],
  "documentationUrls": ["https://docs.aquatech.com/atx900/manual.pdf"],
  "pricing": {
    "supplierId": "SUP-6789",
    "price": 450.00,
    "currency": "AUD",
    "lastUpdated": "2024-05-15T08:00:00Z"
  }
}
```

Infrastructure and Deployment

System Architecture

The Product Knowledge Hub is deployed using a microservices architecture hosted on a private cloud environment compliant with Australian data sovereignty requirements. Key components include:



Scalability & Reliability

- Horizontal scaling of stateless services using Kubernetes orchestration.
- High availability via multi-region deployment within Australian data centers.
- Automated failover and disaster recovery protocols.

Data Backup and Retention

- Regular snapshots of database and document storage.
- Retention policies comply with Australian Privacy Principles (APPs) and corporate governance.

Data Sovereignty and Compliance

The system architecture and operational policies fully observe Australian consumer law, data privacy regulations, and industry best practices:

- **Data Residency:** All customer and product data is stored within Australian jurisdiction.
- **Privacy Compliance:** Personal information is handled in accordance with the Privacy Act 1988 (Cth).
- **Consumer Guarantees:** Product information is factual and non-deceptive, with warranty terms directly sourced from manufacturers.
- **Security Standards:** Conforms to ASD ISM and ISO/IEC 27001 for information security management.

Technical Specifications Summary

Feature	Specification / Standard
Data Transmission	TLS 1.3 (RFC 8446), AES-256 Encryption
Authentication & Authorization	OAuth 2.0, OpenID Connect
Data Storage	PostgreSQL (ACID compliant), Elasticsearch
Compatibility Graph DB	Neo4j (property graph model)
API Format	RESTful API, JSON (RFC 8259)
Search Technology	Apache Lucene / Elasticsearch
Programming Languages	Backend: Java/Kotlin; Frontend: React
Infrastructure	Kubernetes on private cloud within Australia
Security Audits	ISO/IEC 27001, ASD ISM compliance
Data Backup & Retention	Daily incremental, monthly full backups

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Glossary and References

- **AS 3634:** Australian Standard for pool equipment specifications.
- **TLS 1.3:** Transport Layer Security protocol version 1.3, RFC 8446.
- **AES-256:** Advanced Encryption Standard with 256-bit keys.
- **OAuth 2.0:** Open standard for access delegation.
- **OpenID Connect:** Authentication protocol based on OAuth 2.0.
- **ISO/IEC 27001:** International standard for information security management systems.
- **ASD ISM:** Australian Signals Directorate Information Security Manual.
- **Privacy Act 1988 (Cth):** Australian legislation governing privacy and data protection.

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